

Linear Programming Methods for Restless Multi-Armed Bandits: A Survey

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The Restless Multi-Armed Bandit (RMAB)

- **Problem Setup:** A decision maker faces N independent arms, each modeled as a Markov Decision Process (MDP) with two actions: active (1) and passive (0).
- **Restlessness:** Transition kernels P^0 (passive) are not necessarily the identity matrix; arms evolve even when not acted upon.
- **The Curse of Dimensionality:**
 - Joint state space size: d^N (where d is states per arm).
 - Exact solution is **PSPACE-hard** to compute.
- **Goal:** Maximize expected total reward subject to a per-step budget constraint B (often $B = \alpha N$).

The Relaxed Linear Program (RLP)

- **Hard Constraint:** $\sum_{i=1}^N a_i(t) \leq B$ (must hold at every timestep).
- **Relaxation Step:** Replace the hard constraint with a “soft” average constraint:

$$\mathbb{E} \left[\sum_{i=1}^N a_i(t) \right] \leq B \quad \forall t$$

- **The RLP Formulation:**
 - Decouples the arms by optimizing over expected state-action occupancy measures $y_{s,a}(t)$.
 - Constraint: $\sum_s y_{s,1}(t) = \alpha$ (expected proportion of activated arms).

The Fluid Limit: A North Star for Policy Design

- **Concept:** As $N \rightarrow \infty$, the stochastic system approaches a deterministic **Fluid Limit**.
- **The North Star:** The RLP value V_{rel} provides a theoretical upper bound on the achievable reward for any finite N .
- **Convergence:** Policies are designed to “track” the LP solution, aiming for asymptotic optimality where the gap to V_{rel} vanishes as N grows.

Lagrangian Relaxation & The Whittle Index

- **Derivation:** Relax the RLP further by penalizing active actions with a Lagrangian multiplier λ (the “subsidy for passivity”).
- **Index Definition:** The Whittle Index $\lambda(s)$ is the specific subsidy value at which the active and passive actions are equally optimal for state s .
- **Algorithm:** At each step, calculate indices for all arms and pull the top- B arms with the highest indices.

The Condition of Indexability

- **Definition:** An RMAB is indexable if the set of states for which the passive action is optimal increases monotonically with the subsidy λ .
- **The Bottleneck:**
 - Whittle index policies only exist for indexable problems.
 - Indexability is notoriously difficult to prove for many transition kernels and does not hold for all RMAB instances.

Performance and Limitations of Whittle

- **Strengths:** Simple to implement, computationally efficient, and often performs well empirically on independent arms.
- **Failure Modes:**
 - Can fail even in indexable, homogeneous settings when planning horizons are finite or mixing times are large.
 - Multiplicative loss can become arbitrarily large as the discount factor $\gamma \rightarrow 1$ in certain patient engagement models.

Notation: Clustered RMABs

- **Arms and Clusters:** N total arms grouped into K clusters. N_i arms in cluster $i \in [K]$.
- **State and Action Space:** S (states), A (actions), T (planning horizon).
- **Dynamics:** $P_{t,i}(s'|s, a)$ is the transition probability for cluster i at time t .
- **Distributions (Empirical Measures):**
 - $\hat{\mu}_{t,i}(s)$: Actual (random) number of arms in cluster i and state s .
 - $\bar{\mu}_{t,i}(s)$: Deterministic mean-field state (expected counts).
 - $\bar{\alpha}_{t,i}(s, a)$: Mean-field action (expected number of arms receiving action a).
- **Reward/Cost:** $R_{t,i}(s, a)$ and $C_{t,i}(s, a)$.

The Mean-Field MDP Approximation

- **Stochastic System:** The next state $\hat{\mu}_{2,i}(s')$ is a random variable:

$$\hat{\mu}_{2,i}(s') = \sum_{s,a} \sum_{\ell \in [\hat{\alpha}_{1,i}(s,a)]} \mathbb{I}_{\{x_\ell(s,a)=s'\}}$$

- **Mean-Field Limit:** Replace the random variable with its expectation:

$$\bar{\mu}_{2,i}(s') = \mathbb{E}[\hat{\mu}_{2,i}(s')] = \sum_{s \in S, a \in A} P_{1,i}(s'|s, a) \hat{\alpha}_{1,i}(s, a)$$

- **Concentration:** As $N \rightarrow \infty$, the error $\|\hat{\mu}_t - \bar{\mu}_t\|_1$ is typically $O(\sqrt{N})$.
- **Deterministic MDP:** The mean-field state space is a simplex where actions $\bar{\alpha}_t$ can be fractional.

The Mean-Field Linear Program

For the current state $\hat{\mu}_t$, solve the following LP to find the optimal mean-field action $\bar{\alpha}$:

$$\max_{\bar{\alpha}, \bar{\mu}} \sum_{\tau=t}^T \sum_{i,s,a} \gamma^{\tau-1} \bar{\alpha}_{\tau,i}(s,a) R_{\tau,i}(s,a)$$

Constraints:

- **Initial Condition:** $\bar{\mu}_{t,i} = \hat{\mu}_{t,i}$ (using current real-life state).
- **State Evolution:** $\bar{\mu}_{\tau+1,i}(s') = \sum_{s,a} P_{\tau,i}(s'|s,a) \bar{\alpha}_{\tau,i}(s,a)$.
- **Consistency:** $\sum_{a \in A} \bar{\alpha}_{\tau,i}(s,a) = \bar{\mu}_{\tau,i}(s) \quad \forall \tau, i, s$.
- **Resource/Budget:** $\sum_{i,s,a} \bar{\alpha}_{\tau,i}(s,a) C_{\tau,i}(s,a) \leq B_{\tau} \quad \forall \tau$.

The Mean-Field Linear Program

$$\max_{\bar{\alpha}_t, \bar{\mu}_t} \sum_{t,i,a} \gamma^{t-1} \bar{\alpha}_{t,i}(\cdot, a)^\top R_{t,i}(\cdot, a)$$

$$\text{s.t.} \quad \bar{\mu}_{1,i} = \hat{\mu}_{1,i}, \quad \forall i$$

$$\bar{\mu}_{t+1,i}(\cdot)^\top = \sum_{a \in A} \bar{\alpha}_{t,i}(\cdot, a)^\top P_{t,i}(\cdot | \cdot, a), \quad \forall t \in [T-1], \forall i$$

$$\sum_{i \in [K]} \sum_{a \in A} \bar{\alpha}_{t,i}(\cdot, a)^\top C_{t,i}(\cdot, a) \leq B_t, \quad \forall t$$

$$\sum_{a \in A} \bar{\alpha}_{t,i}(\cdot, a) = \bar{\mu}_{t,i}(\cdot), \quad \forall t, i$$

$$\bar{\alpha}_{t,i}(\cdot, a) \geq 0, \quad \forall t, i, a$$

The MFP Algorithm: Solve and Round

For each decision epoch $t = 1, \dots, T$:

- ➊ **Observe:** Measure the current real-life state counts $\hat{\mu}_t$.
- ➋ **Solve:** Run the Mean-Field LP with $\hat{\mu}_t$ as the starting state and horizon $(T - t + 1)$.
- ➌ **Quantize:** Since $\bar{\alpha}_{t,i}(s, a)$ may be fractional, round down:
 - Play action a for $\lfloor \bar{\alpha}_{t,i}(s, a) \rfloor$ arms in cluster i , state s .
- ➍ **Budget Safety:** For any remaining unassigned arms:
 - Play the zero-cost action.
 - This guarantees the implementation stays within budget B_t .
- ➎ **Recede:** Observe the realization $\hat{\mu}_{t+1}$ and repeat.

Advantages of the Mean-Field Policy

- **Assumption-Free:** Does not require indexability, homogeneity, or global attractor properties.
- **Scalability:** Handles thousands of arms effectively by solving small LPs relative to cluster/state counts.
- **Error Correction:** Solving the LP at **every** time step corrects for:
 - **Rounding Error:** Introduced by flooring the actions.
 - **Variance Error:** Introduced by stochastic transitions.
- **Near-Optimality:** Provably $O(\sqrt{K|S|N})$ sub-optimal with high-probability confidence bounds.

The CORMAB Challenge: Coupled Actions

- **Beyond top-K:** Traditional RMABs assume arm actions are independent.
- **CORMAB (Coupled RMAB):** Actions are constrained across arms due to combinatorial requirements.
- **Examples:**
 - **Path Planning:** Decision to pull arm A depends on the physical distance/route from arm B .
 - **Bipartite Matching:** Resource assignment constraints.
 - **Nonlinear Compounding:** Multiple interventions impacting the same arm simultaneously.

SEQUOIA Architecture: Q-Net in a MILP

- **Problem:** Action spaces are exponentially large; top- K selection is impossible.
- **Solution: Embed a trained Q-network** directly into a Mixed-Integer Linear Program (MILP).
- **ReLU Embedding:** ReLU units are modeled using binary variables and **big-M constraints**:

$$w^T y + b = x - s, \quad x \geq 0, s \geq 0, \quad z \in \{0, 1\}$$

$$z = 1 \Rightarrow x \leq 0; \quad z = 0 \Rightarrow s \leq 0$$

Performance in Combinatorial Action Spaces

- **Result:** SEQUOIA outperforms existing RL methods (which cannot handle combinatorial selection) by an average of **24.8%** on CORMAB instances.
- **Sequential vs. Myopic:** Significantly beats myopic baselines by learning the long-term value via DQN updates, despite the combinatorial bottleneck.
- **Efficiency:** Uses memoization of MILP solutions and off-policy warm-starting to handle the high cost of repeated optimization solves.

Method Comparison Summary

Feature	Whittle Index	LP-Update/MF	SEQUOIA
Scalability	High (N)	Medium (LP size)	Low (MILP size)
Constraints	Simple top- K	Linear Budget	General Combinatorial
Optimality	Asymptotic	Asymptotic (Exp. possible)	Learned (Empirical)
Assumptions	Indexability required	No indexability	No independence
Complexity	$O(N \log N)$	Polynomial (LP)	NP-hard (MILP)

- **Whittle** is best for massive N and independent arms.
- **LP-Update** is best for medium-to-large N without structural proofs.
- **SEQUOIA** is necessary for complex, strongly-coupled constraints.

- **Linear Programming** provides the necessary relaxation to handle the state-space explosion of RMABs.
- **Classical Methods** (Whittle) are simple but fragile in practical finite-horizon or non-indexable settings.
- **Modern Methods** (LP-Update, Mean-Field) provide stronger guarantees (exponential convergence) without structural assumptions.
- **Future Frontiers** (SEQUOIA) integrate deep learning with MILP to bridge the gap between sequential planning and hard combinatorial constraints.